



Emerging AI threats in cybercrime: a review of zero-day attacks via machine, deep, and federated learning

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Abstract

The rise of artificial intelligence (AI) revolutionized both cybersecurity defenses and cybercriminals' methods to exploit vulnerabilities. Cybercriminals continue to exploit previously undiscovered vulnerabilities, known as zero-day attacks, posing severe threats to cybersecurity. These attacks are particularly challenging to detect, as they target unknown weaknesses in systems before security teams can respond or act. Traditional intrusion detection systems (IDS) rely heavily on pre-existing attack signatures, making them ineffective against zero-day threats. Machine learning (ML) algorithms have recently become a promising solution for enhancing IDS capabilities by identifying anomalies and predicting potential vulnerabilities in real time. This review paper explores how cutting-edge AI techniques, specifically ML, DL, and federated learning (FL), are harnessed to counter zero-day attacks. AI is used to defend against cyberattacks that exploit vulnerabilities unknown to existing security software. This research explores different AI methods used in cybersecurity, analyzes the data used to train these AI models, and evaluates how well various algorithms perform in actual cyberattacks. Moreover, key challenges in deploying ML for zero-day detection are highlighted, including handling imbalanced data, generalization across diverse types of attacks, and the trade-offs between accuracy and computational cost. The paper outlines future research directions to enhance AI-based zero-day attack defenses and strengthen proactive cybersecurity strategies.

Keywords Machine learning · Cybersecurity · Federated learning · Zero-day attacks

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Abbreviations

AI	Artificial intelligence
IDS	Intrusion detection system
ML	Machine learning
FL	Federated learning
DoS	Denial of service
DDoS	Distributed denial of service
APT	Advanced persistent threat
IPS	Intrusion prevention system
PFL	Personalized federated learning
IID	Independent and identically distributed
SQL	Structured query language
IoT	Internet of things
NLP	Natural language processing
SIDS	Signature-based intrusion detection system
AIDS	Anomaly-based intrusion detection system
NN	Neural network
RNN	Recurrent neural network
CNN	Convolutional neural network
P	Precision
R	Recall
F	F-measure
FNR	False negative rate
FPR	False positive rate
TP	True positive
FP	False positive
TN	True negative
FN	False negative
MCC	Matthews correlation coefficient
ROC-AUC	Receiver operating characteristic / Area under curve
CIC	Canadian institute for cybersecurity
AWS	Amazon web services
DoH	DNS over HTTPS
SAE	Stacked autoencoders
HBO	Honey badger optimization
NGO	Northern Goshawk optimization
NFV	Network function virtualization
SSCL	Self-supervised contrastive learning
CBTB	Communication-based train control
NAR	Neural algorithmic reasoning
ELM	Extreme learning machine
RF	Random forest
DT	Decision tree
KNN	K-nearest neighbors
NB	Naïve Bayes
ITS	Intelligent transportation system
IoMT	Internet of medical things

H-FL	Hierarchical federated learning
FTL	Federated transfer learning
XAI	Explainable artificial intelligence
LIME	Local interpretable model-agnostic explanations
SHAP	SHapley additive exPlanations
PLR	Penultimate layer representation
CKA	Centered kernel alignment
ZTA	Zero trust architecture

1 Introduction

In the recent digital era, mitigating cyber risks is crucial for organizations, governments, and individuals [1]. The proliferation of digital systems has coincided with a rise in the volume and sophistication of cyber threats, making it imperative to implement advanced defense mechanisms [2, 3]. Cybercriminals continuously seek out vulnerabilities in systems and networks to execute malicious activities, ranging from stealing sensitive data to disrupting critical infrastructure [4, 5]. Zero-day attacks are among the most dangerous threats, which exploit previously unknown vulnerabilities, giving defenders no time to react before the attack begins [6, 7]. As a result, securing information systems has become a fundamental aspect of modern cybersecurity strategies [8].

Cybersecurity threats come in many forms, as shown in Fig. 1, each with varying complexity and potential damage [9]. Malware attacks, including viruses, worms, and ransomware, can compromise millions of devices worldwide [10]. Phishing tricks people into sharing personal data, while Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS) attacks overload networks, causing service disruptions [11, 12]. More sophisticated methods, such as Advanced Persistent Threats (APTs), involve prolonged infiltration into a target network, often for espionage or sabotage [13–15]. Traditional security tools, such as firewalls and antivirus software, are becoming insufficient in the face of increasingly sophisticated and voluminous cyberattacks, especially zero-day threats that target unpatched flaws [16]. Recent attacks utilize undiscovered vulnerabilities and advanced techniques to evade detection systems [17].

Given the evolving nature of cyber threats, IDS and intrusion prevention systems (IPS) are crucial for detecting malicious activities within networks [18]. An IDS monitors traffic, analyzing incoming and outgoing data to detect potential threats (malicious activities) [19, 20]. Conventional IDSs, which rely on predefined signatures of known attacks, are highly effective in catching familiar threats but struggle against novel or previously unseen attacks like zero-day exploits [21]. The limitations of signature-based detection have driven the adoption of more sophisticated detection techniques, especially those that leverage ML [22].

ML is a branch of AI that empowers systems to acquire knowledge and enhance performance through experience without explicit coding. [23–27]. In cybersecurity, ML models play a key role in flagging abnormal behavior and detecting anomalies that may signal impending or ongoing attacks [28]. Unlike traditional methods, ML-based models can recognize subtle patterns in data, even when faced with previously unknown attack vectors [23, 29, 30]. Various ML approaches are used in cybersecurity, including supervised learning, where models learn from labeled data to make predictions, and unsupervised learning, which focuses on detecting unusual patterns or anomalies without predefined labels [31]. Reinforcement learning methods dynamically adapt to changing threats by learning from past actions [32].

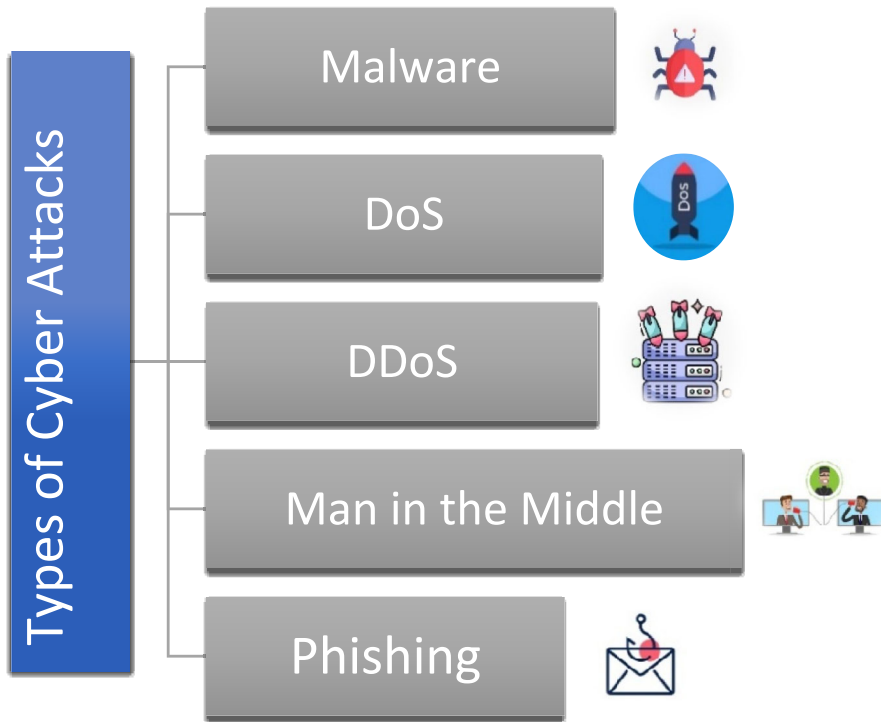


Fig. 1 Types of cyberattacks

The importance of ML in cybersecurity cannot be overstated [33, 34]. ML algorithms enhance the capabilities of IDS to identify and mitigate novel attacks [35]. These models can detect even subtle anomalies of a cyberattack by continuously analyzing vast amounts of network and system data. Moreover, ML models can adapt to changing attack patterns, making them more flexible and capable of detecting zero-day vulnerabilities that would go unnoticed by traditional IDS solutions. As cyberattacks evolve, the defenses and ML-based systems must be at the forefront of this battle [36].

However, as cyber threats become more complex, challenges arise in scaling ML solutions across distributed networks with varied data sources. FL is a promising approach to overcome these challenges [37, 38]. FL is a decentralized ML approach that enables individual nodes to collaboratively train with a shared model without sharing raw data. FL, a collaborative ML technique, was introduced by Google in 2016 [39–41]. This is particularly important in cybersecurity, where privacy is concerned and the need for real-time threat detection makes traditional centralized learning impractical [42]. FL keeps data localized to each device while sharing only the model updates, ensuring data privacy and collaboration [43–45].

The potential of FL to revolutionize cybersecurity is substantial, although challenges remain in its deployment. Enabling distributed learning, FL can enhance detection models, especially for zero-day attacks across different environments. With personalized federated learning (PFL), individual models can be fine-tuned to address the unique threat landscape of each client, making it a more robust solution in non-IID (independent and identically

distributed) environments, where data distributions differ from one device or node to another [46, 47].

This review paper comprehensively reviews ML, DL, and FL algorithms and their application in detecting zero-day attacks within cybersecurity. Various models and datasets that derive their performance are analyzed. The emerging role of FL in cybersecurity is also discussed. By examining advancements in ML-based IDS and addressing the remaining challenges, this review offers a foundation for future research to enhance defenses against evolving cyber threats.

The selection process for this survey relied on a comprehensive search of well-established academic databases, such as Google Scholar, IEEE Xplore, Science Direct, ResearchGate, Springer, ACM Digital Library, Scopus, Web of Science, Taylor and Francis, and Inder-science. Our scope was defined by objectives targeting zero-day attack detection using AI techniques, particularly ML, DL, and FL. The inclusion criteria required studies to focus on AI-driven methods for zero-day attacks relevant to our themes in peer-reviewed journals or reputable conferences.

The study was conducted using Google Scholar, IEEE Xplore, Science Direct, ResearchGate, Springer, ACM Digital Library, Scopus, Web of Science, Taylor and Francis, and Inderscience, with keywords like ‘zero-day attacks AND machine learning’, ‘deep learning in cybersecurity’, and ‘federated learning for zero-day threats’. Results were screened by titles and abstracts to identify relevant studies, followed by a full-text review to ensure alignment with our research focus. Additionally, data points such as model types, datasets, and evaluation results were extracted and analyzed to identify trends and gaps in the literature review.

To the best of our knowledge, this literature review stands out by focusing on recent studies from 2024 and integrating various AI techniques: ML, DL, and FL. As a result, this review provides a robust comparison of cyberattack frameworks, offering valuable insights for researchers and serving as a research-based guide. Additionally, it emphasizes the latest scientific papers and highlights cutting-edge advancements in the field.

The remainder of this survey is organized as follows: Sect. 2 introduces a background of zero-day attacks, IDS, and enhancing IDSs and Zero-day attack detection by AI techniques. Section 3 focuses on the benchmark datasets for IDS and zero-day attack detection. Section 4 discusses data preprocessing and feature selection for IDS and zero-day attack detection. Section 5 focuses on zero-day attack detection based on AI techniques. Finally, Sects. 6 and 7 provide future directions and conclusions.

2 Background

Cybersecurity is the process by which any organization tries to protect its systems that are interconnected and exposed to the Internet from various cyber threats [5]. This practice is a common security measure, used by individuals and enterprises to shield their data centers and servers from unauthorized access [37]. The best practice defines infosec risk management within the scope of assets, threats, vulnerabilities, and events [48]. Figure 2 indicates how the attackers exploit the assets [49, 50]. Zero-day attack detection solutions aim to proactively detect unknown threats targeting valuable assets within a given system [51].

Cybersecurity also aims to prevent external attacks on sensitive data. A cyberattack is an attack that an individual or organization breaches information to gain some benefit from disrupting the victim’s system. Cyberattacks hit businesses every day, and cybercrime has also

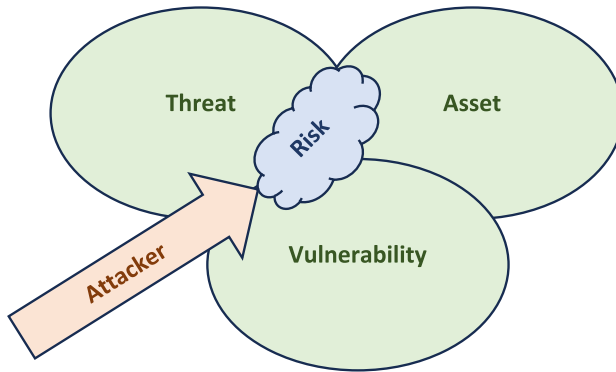


Fig. 2 How the attackers exploit the assets

increased every year [52]. The CIA triad model [36] sets the foundation for all security-based frameworks, which have been used for the past two decades to enforce security policies. It identifies possible threats and necessary solutions for those threats in the field of information security. The CIA triad model evolves around three major components [53, 54], as shown in Fig. 3.

Cyberattacks vary widely and can be classified according to the attackers' intentions and the information they need to gain access [55, 56]. Malware is malicious software that activates when the user clicks an unauthorized link or email attachment [10]. Phishing is a common cyberattack where attackers send deceptive emails pretending to be legitimate sources to trick people into revealing sensitive information [57]. Man-in-the-middle is another interesting



Fig. 3 The CIA triad model

attack where the intruders target the network itself. So, the sender will be sending the message without knowing that the hacker in the middle is receiving the information [58, 59]. One of the popular attacks, DoS, floods the system and the server with traffic to exploit the available resources and bandwidth. Unfortunately, the legitimate request cannot be addressed due to DoS attacks [60]. Structured Query Language (SQL) attack inserts the malicious code into the server and forces the server to reveal sensitive information from the database [61, 62]. A Zero-day (0-day) attack is an unknown security threat in software applications for which the patch has not been released, or software developers are unaware of the attack [7, 16]. When such an attack occurs for the first time, defenders are given zero days of notice (hence the name) [62].

With the evolving digital era, cyberattacks have grown in volume and complexity not only against single systems but also against critical infrastructures. DDoS attacks are among the most prevalent threats whose goal is to disrupt service availability by overwhelming systems with traffic. Volumetric DDoS attacks are type that engulfs networks with extremely large volumes of traffic, consuming all available bandwidth and rendering services inaccessible [63]. These attacks have taken center stage due to the prevalence of Internet of Things (IoT) devices and the growth in botnet-fueled amplification tools, a major menace to current detection and countermeasures [63].

Worldwide DDoS attacks, both protocol and application-layer attacks, are becoming increasingly advanced through the utilization of high-level evasions. Their evading capabilities usually allow them to get beyond conventional security filters; therefore, precise and timely detection is increasingly necessary [64]. Botnet attacks are one of the main drivers of such threats. A botnet consists of compromised systems owned by an attacker, used not only to launch DDoS attacks but also to deliver spam, credential stuffing, and spreading malware. Present studies emphasize the requirement of ML models for botnet detection with low false positives, and scalability [65].

Malware assaults are one of the most potent and versatile forms of cyberattacks. Variants like ransomware, spyware, and trojans can also steal sensitive information, encrypt files for ransom, or provide remote control to attackers. Malware's obfuscated and dynamic characteristics make its detection using traditional signature-based methods useless, hence the shift toward AI-based behavioral analysis methods [66]. New, polymorphic, or zero-day malware variants must be caught by models that learn from previously observed patterns and do not overfit.

Another prevalent threat is phishing, which uses impersonating communication (typically websites or emails) to encourage users to give away credentials or install dangerous payloads. Phishing contributes toward a high success rate of successful data breaches and is typically used as a means of entry into broader intrusions. Phishing detection has recently explored the application of Natural Language Processing (NLP) and DL methods in detecting fine-grained signals in content, structure, and purpose [67]. Fighting so many forms of cyberattacks, each with unique data characteristics and behavior traits refers to the need for adaptive and intelligent AI-powered systems, especially in handling unknown or zero-day variants.

2.1 Zero-day attacks

A zero-day attack is a new type of cyberattack that uses vulnerabilities that have not been publicly identified [68]. These attacks are often undetected because they use new methods that avoid detection by existing security tools [69]. This allows attackers to target specific systems without being noticed [23, 70]. Bilge and Dumitras [70] studied how long these

Table 1 Zero-day terminologies

Zero-day Vulnerability	A zero-day vulnerability is one in which no fix is promptly available, and the seller could conceivably know. The non-attendance of a fix for multi-day helplessness introduces a zero-day risk to sellers and clients [23]
Zero-day Exploit	A zero-day exploit is a type with a bit of programming, information, or a set of directions to exploit zero-day weakness [23, 73]
Zero-day Attack	A zero-day attack abuses zero-day helplessness [23]

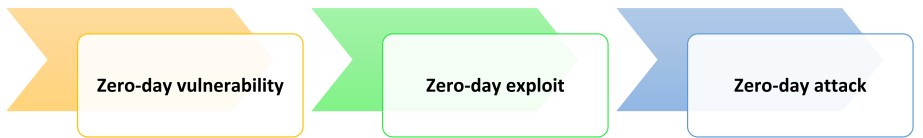


Fig. 4 Zero-day attack workflow

attacks last and how often they occur. Their findings show that a typical zero-day attack remains undetected for about 312 days on average. These attacks can rise significantly once vulnerabilities are made public [23, 69].

A zero-day vulnerability is a computer software vulnerability that is unknown to those who would be interested in mitigating the vulnerability. Until the vulnerability is mitigated, hackers can exploit it to adversely affect computer programs, data, additional computers, or a network [71]. The security vulnerability in the software is susceptible to malware infection, remains unknown, and unfixed by the vendor, which is further exploited by malicious cyber criminals even before the vendor becomes aware of it [72]. This type of exploit is called a zero-day attack [23]. Table 1 defines the important terminologies related to zero-day, and Fig. 4 shows the workflow [23].

The primary difference between zero-day attacks and n-day attacks lies in the stage of vulnerability exploitation [74]. Zero-day attacks target vulnerabilities are entirely unknown to the vendor and public, meaning no patch or fix exists when the attack occurs [75, 76]. This makes zero-day attacks particularly dangerous, as defenders have no prior knowledge or defense mechanisms, and the attack can go undetected for some time [75, 77]. N-day attacks are the exploitation of vulnerabilities that have already been disclosed and, in many cases, have been patched [78]. However, attackers in n-day scenarios take advantage of systems that have not yet applied these patches, relying on delayed patch management [79]. While n-day attacks can be mitigated by timely updates, zero-day attacks leave little room for immediate defense, requiring a more reactive approach [80]. In essence, zero-day attacks represent the exploitation of newly discovered weaknesses, whereas n-day attacks exploit known vulnerabilities that have not been adequately addressed [81]. Table 2 summarizes these differences between 0-day attacks and n-day attacks.

2.2 Intrusion detection systems

The first IDS was proposed in 1980 [82, 83]. For an IDS, an intrusion means an attempt to access information about computer systems or to damage the operation of the system in an illegal or unauthorized manner [84]. An IDS is a computer security application that aims to detect various security violations, from attempted break-ins by outsiders to system

Table 2 Zero-day attacks vs N-day attacks

Criteria	Zero-Day Attack	N-Day Attack
Definition	Exploits vulnerabilities unknown to the vendor or public, with no patch available	Exploits vulnerabilities that have been publicly disclosed, and a patch or mitigation is available
Discovery Stage	It occurs before a vulnerability is made public or patched	Occurs after the vulnerability is disclosed and possibly patched
Exploitation Window	Potentially high, as defenders are unaware of the vulnerability	Limited because patches or fixes are available; attackers exploit delayed patching by users
Risk Level	Very high due to no immediate defense mechanisms	Moderate to high, depending on how quickly patches are applied
Difficulty to Detect	It is difficult to detect since no known signatures exist in antivirus or security systems	It is easier to detect with known signatures and patches already available
Attacker Advantage	The attacker has a significant advantage, as defenses are typically non-existent	Attackers must exploit systems that have not yet applied the available patches
Mitigation	It is impossible to mitigate immediately; defense is often reactive	It can be mitigated by applying patches or updating systems regularly
Impact on Organizations	High impact, particularly on critical infrastructure or high-value targets	The impact varies; it can be high if patches are not applied promptly
Examples	Stuxnet (2010), EternalBlue exploit (2017)	Exploitation of Microsoft Exchange Server vulnerabilities (post-patch delay) in 2021

penetrations and abuses by insiders [85, 86]. The main functions of IDSs are to monitor hosts and networks, analyze the behaviors of computer systems, generate alerts, and respond to suspicious behaviors [83, 87]. Because they monitor related hosts and networks, IDSs are typically deployed near the protected network nodes (e.g., the switches in major network segments) [83].

The components of IDS, as depicted in Fig. 5, typically include sensors, which collect data, a detection engine, which analyzes the data for signs of malicious activity, and a response component, which acts when an intrusion is detected [88].

It is worth mentioning that IDS can be categorized into two main classes: signature-based intrusion detection systems (SIDS) and anomaly-based intrusion detection systems (AIDS) [18, 89]:

- **Signature-based intrusion detection systems:**
They are designed to detect and raise the alarm when a network's traffic or host events match one of the signatures [83]. However, the repository of signatures needs to be updated regularly to include new attacks and remove no longer efficient rules [18, 84, 90]. SIDSs are also named misuse-IDSs [91]. SIDSs are efficient in detecting known attacks, but they fail to identify novel attacks [88].
- **Anomaly-based Intrusion Detection Systems:**
They are based on a network's predefined behavior; when a new event occurs, it is accepted, or anomaly detection is triggered [92]. The recognized network behavior is developed or acquired based on the network administrators' specifications [84, 90, 93]. AIDs can detect

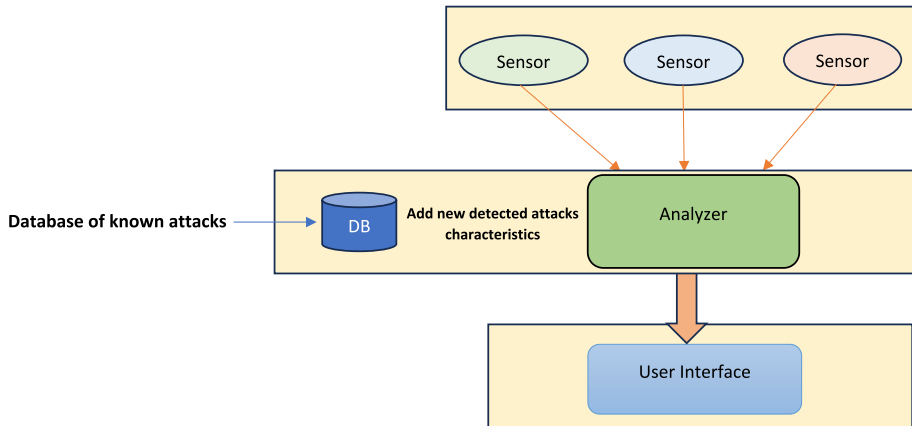


Fig. 5 IDS general architecture

new attacks successfully due to their ability to identify deviations from normal profiles [94].

2.3 Enhancing IDSs and Zero-day attack detection by AI techniques

This section explores the enhancement of IDSs through advanced AI techniques, aiming to improve the detection of zero-day attacks and other sophisticated cyber threats. The discussion will cover how AI-based IDSs outperform traditional systems by leveraging ML, DL, and anomaly detection methods to recognize new and evolving attack patterns.

2.3.1 Machine learning

ML is the core field of AI that utilizes statistical techniques to enable systems to learn from data and make predictions or decisions without requiring explicit coding [95]. The primary objective is to create autonomous systems that can learn from experience and adapt to changing environments [23, 96, 97].

Supervised ML is an ML paradigm that employs labeled examples to train models that can predict future occurrences [98]. The learning algorithm learns a mapping function from input features to output labels by minimizing a loss function on the training data. This learned model can then be used to make predictions on new, unseen data [23, 69, 99, 100].

Unsupervised ML is a ML paradigm that allows systems to identify hidden structures and relationships within unlabeled datasets. Unsupervised learning techniques, such as clustering and dimensionality reduction, can be used to explore and visualize large datasets [23, 100, 101].

Semi-supervised ML is a hybrid approach combining supervised and unsupervised learning elements. It significantly enhances learning accuracy, particularly when labeled data are limited and costly to acquire [23, 97].

Reinforcement ML is an ML paradigm that involves an agent interacting with an environment, receiving rewards or punishments for its actions, and learning a policy that maximizes the expected cumulative reward [23, 68, 101–103].

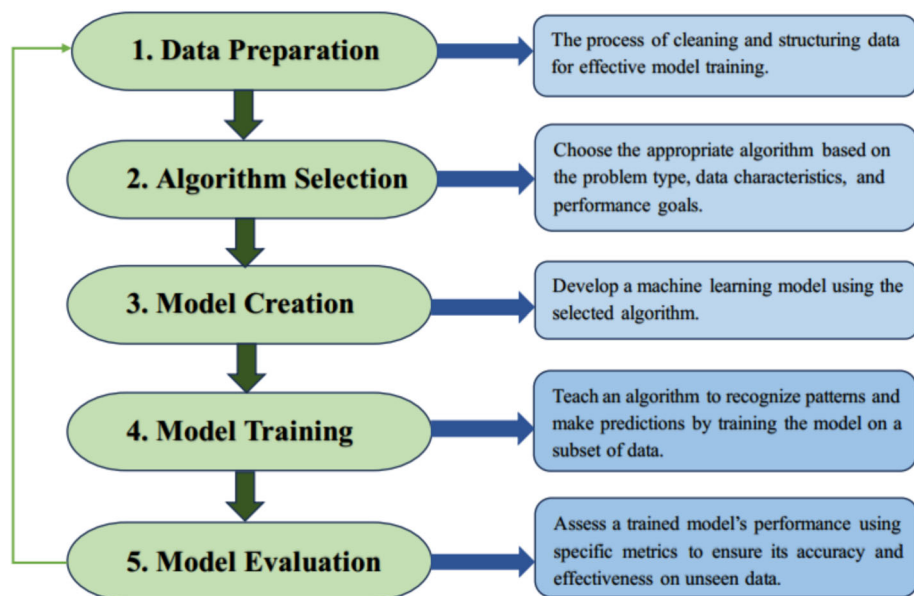


Fig. 6 The ML Process

To build an ML model, one must first gather and prepare relevant data, select an appropriate algorithm, partition the dataset into training and testing subsets, train the model on the former, and evaluate its performance on the rest [23]. These steps are shown in Fig. 6.

2.3.2 Deep learning

DL, a subset of AI, has emerged as a powerful tool for cybersecurity. Neural network (NN) architectures, such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs), have been employed to address challenges effectively, like intrusion detection and malware classification. These models have the potential to outperform traditional ML techniques in identifying APTs by analyzing temporal network traffic patterns [104].

ML techniques can significantly enhance the performance of cybersecurity systems by improving anomaly detection and threat classification [105]. ML algorithms can process vast amounts of data to identify patterns indicative of malicious activities [106]. ML models, leveraging supervised, unsupervised, and DL techniques, can effectively classify and categorize various cyber threats, including malware, phishing attacks, and network intrusions. These models can continuously learn from new data, improving their ability to detect and respond to emerging threats [107, 108]. Integrating ML into cybersecurity significantly enhances both the accuracy and efficiency of threat detection by empowering systems to analyze vast quantities of data, identify intricate patterns, and minimize false positives. Furthermore, ML models, particularly DL techniques such as CNNs, facilitate the development of proactive defense mechanisms that can adapt to evolving threats, thereby providing a dynamic and robust approach to cybersecurity [109].

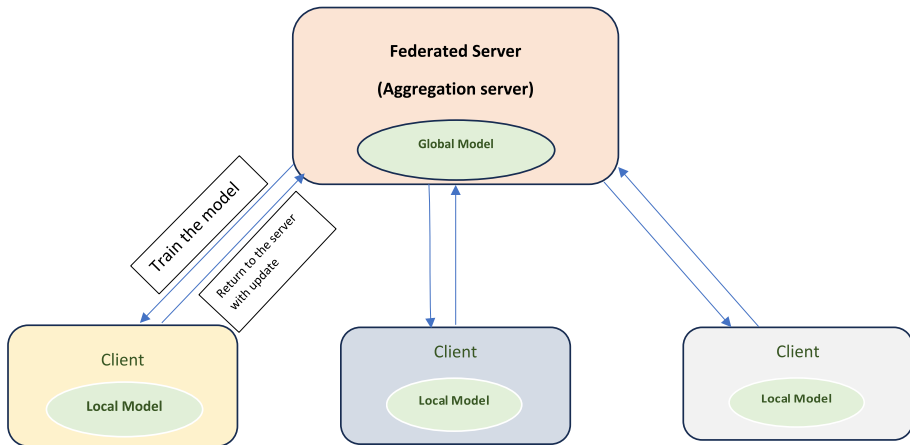


Fig. 7 The Concept of FL

2.3.3 Federated learning

Google was the first to introduce the concept of FL by McMahan et al. in 2016 [110–112] as a distributed [113] and collaborative [114] approach to ML. FL is a novel ML/DL approach [115, 116] in which a set of entities is trained over their local data to build a global model [110], as illustrated in Fig. 7. It is a decentralized ML approach that allows multiple clients to collaboratively train a shared model while preserving data privacy and security [112, 113].

Unlike traditional ML approaches, end nodes do not need to share their data with typical data centers to be further processed [117]. FL has emerged as a promising approach due to its superior advantages over centralized learning techniques [118]. FL addresses the challenges of data privacy and security by enabling collaborative model training without sharing raw data [119] due to its enhanced security and privacy features [120]. As shown in Table 3, there are differences between traditional centralized systems and FL. Traditional ML and DL paradigms are confronted with significant limitations, including privacy concerns, scalability issues, computational inefficiency, and latency bottlenecks [39]. FL, a novel and innovative AI paradigm, presents a viable solution to the privacy, scalability, and efficiency challenges associated with traditional ML and DL approaches [111]. FL enables multiple clients to participate in model training while preserving data privacy through secure aggregation of model updates [7]. Table 3 shows the main differences between centralized learning and FL.

2.3.4 Evaluation metrics

Multiple metrics are utilized to evaluate the performance of ML models in IDSs. These metrics are used to select optimal models. To comprehensively measure detection efficacy, a combination of metrics is often employed [83, 110, 121–123].

- Accuracy is a suitable metric for balanced datasets, where the number of samples in each class is approximately equal. However, in imbalanced datasets, where one class significantly outnumbers the other, accuracy can be misleading, as the classifier may achieve

Table 3 Centralized learning vs federated learning

Criteria	Centralized learning	Federated learning
Data Location	Collected and stored in a centralized server	Distributed across multiple devices or sources
Data Privacy	Data must be transferred to the central server	Data remain on the local device
Model Training	Training occurs on a centralized server	Training occurs locally on devices
Communication Overhead	Lower, as data are already centralized	Higher due to the need to transmit model updates
Scalability	Limited by server capacity and network infrastructure	Highly scalable due to distributed data processing
Risk of Data Breach	Higher, due to the central storage of sensitive data	Lower, as raw data are not transmitted
Latency	Lower, as all processes are centralized	It can be high due to communication between devices Potential delays in communication between devices in decentralized systems, affecting real-time detection
Fault Tolerance	Central server failures can disrupt the entire system, while local devices may not function independently	More resilient, local devices can still function independently
Regulatory Compliance	Managing compliance with data privacy regulations is more complex with centralized data storage	Easier to comply with data privacy regulations

high accuracy by simply predicting the majority class [110].

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}} \quad (1)$$

- **Precision (P)** is a key performance metric for imbalanced classification problems, as it measures the model's ability to identify positive instances accurately while minimizing false positive classifications [121].

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

- **Recall (R)** is a crucial metric for evaluating IDS performance, as it measures the model's ability to identify attacks accurately and minimize false negatives [121].

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3)$$

- **F-measure (F)** is a valuable metric for evaluating the performance of classification models, particularly in imbalanced datasets, as it considers both P and R [122].

$$F = \frac{2 * P * R}{P + R} \quad (4)$$

- **The false negative rate (FNR)** is a critical metric for evaluating IDS performance, as it measures the model's ability to correctly identify attacks and avoid false negatives [123].

$$\text{FNR} = \frac{\text{FN}}{\text{TP} + \text{FN}} \quad (5)$$

- **The false positive rate (FPR)** is a critical metric for evaluating the performance of IDS, as it measures the model's ability to minimize false alarms and avoid unnecessary alerts [123], and it is calculated as follows:

$$\text{FPR} = \frac{\text{FP}}{\text{TP} + \text{FP}} \quad (6)$$

True positive (TP), false positive (FP), true negative (TN), and false negative (FN) are key performance metrics used to evaluate the performance of IDS. In attack detection, the positive class represents malicious activity, and the negative class represents benign activity. The selection of appropriate performance metrics depends on the specific goals of the IDS, such as maximizing detection accuracy, minimizing false alarms, or balancing both [22].

- **The Matthews Correlation Coefficient (MCC)** is a comprehensive metric for evaluating the performance of binary classification models. All four confusion matrix components (TP, TN, FP, and FN) are integrated to form an equal appraisal of the model's quality. MCC is particularly useful in cases with imbalanced datasets because it does not consider class distribution and does not get skewed toward the dominant class [124]. MCC measures the degree of divergence between the actual labels and the predicted label. Its range is from -1 to $+1$, with $+1$ representing flawless forecasting accuracy, 0 representing equal chance performance, and -1 representing perfect inverse correlation, which showcases total divergence between predictions and outcomes [125]. Many researchers emphasize the role of MCC when measuring the impact of IDS [126].

$$\text{MCC} = \frac{(\text{TP} * \text{TN}) - (\text{FP} * \text{FN})}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \quad (7)$$

- **Area Under the Receiver Operating Characteristic Curve (ROC-AUC)** is widely used as an evaluation metric, mostly in binary classification problems. It measures the area under the ROC curve, which represents the relationship between the true positive rate and FPR at different levels of the classifier. The AUC provides a holistic assessment of how well a model performs at all possible decision thresholds. AUC of 0.5 suggests no discrimination ability, identical to random chance, and 1 indicates perfect classification [124].
- **Cohen's Kappa** is a statistical measure that assesses the relationship between the predicted and actual class outcomes while accounting for random chance agreement. It is quite useful for evaluating classification models where inter-rater or predictor reliability is important. The Kappa coefficient can range from -1 to $+1$. Values close to $+1$ represent strong agreement, 0 represents an agreement that is no better than stochastic chance, while negative indicates a level of agreement that is less than random agreement [124, 127].

3 Benchmark datasets for IDS and zero-day attack detection

Several benchmark datasets are pivotal in assessing IDS and the detection of zero-day attacks. The CICIDS2017 dataset [128], created by the Canadian Institute for Cybersecurity (CIC) in 2017, provides extensive records of contemporary internal and external cyber incidents

[129]. This dataset, featuring five days of benign and malicious traffic data, includes PCAP recordings. It has been preprocessed for ease of use by researchers, making it ideal for evaluating IDS performance [129, 130]. In comparison, KDDCUP'99 [114, 115], a widely utilized dataset, has highlighted limitations such as uneven class distribution and redundant data that can impair anomaly detection assessments [83, 131]. The NSL-KDD dataset [132, 133], released by CIC in 2009, addresses these issues by balancing class sizes and eliminating duplicate entries [130, 131]. This improved dataset supports evaluations of four primary attack types: DoS, U2R, R2L, and probing attacks [130].

Additional datasets also contribute to IDS and anomaly detection research. UNSW-NB15 [134] captures realistic network traffic from a university environment and is suited for unsupervised learning models, such as Isolation Forest, that distinguish between normal and anomalous behaviors [18]. The CIC-AWS-2018 dataset [135], collected from the Amazon Web Services (AWS) cloud, is tailored to IDS assessments in cloud computing scenarios [18, 136]. Similarly, CIRA-CIC-DoHBrw-2020 [137] captures both benign and malicious DNS over HTTPS (DoH) traffic alongside non-DoH data, enhancing IDS performance analysis in encrypted traffic environments [18, 138]. Each of these datasets presents unique characteristics, as summarized in Table 4, facilitating an informed selection for research and development in IDS.

As it appears in the table, there are four common attacks: DoS, Probe, R2L, and U2R [131, 140], the descriptions of them shown in the following table (Table 5).

4 Data Preprocessing and Feature Selection for IDS and Zero-day Attack Detection

In the field of IDS designed especially to detect zero-day attacks, data preprocessing, and feature selection play an important part in ensuring the model's accuracy and computational efficiency. Effective data preprocessing is an important step in building stable ML models for intrusion detection and recognition of zero-day attacks because model performance is derived from input data quality. Preprocessing begins with cleansing steps that address missing values, eliminate noise, and fix errors to provide accuracy and consistency in the data. Normalization is then applied to normalize numerical features to a consistent range, an important step that enhances algorithmic efficiency and convergence [144]. Dataset balancing also decreases bias toward majority classes, hence the performance of the model in detecting rare or novel attack patterns [145]. Since many ML algorithms require numerical inputs, categorical variables are encoded into suitable numerical representations to enhance compatibility and optimal learning [145]. Handling missing values is necessary by imputing or removing missing values to prevent errors during model training. Effective handling of missing data ensures the integrity of the dataset [146]. Collectively, these preprocessing stages are the building blocks of effective and robust IDS that can identify even novel types of threats.

Feature selection is important in constructing effective and efficient IDS, especially for detecting zero-day threats. Stacked autoencoders (SAEs) are one of the well-known methods for unsupervised feature extraction, widely used in DL models. These DL models can reduce data dimensionality without compromising on the important structures and relationships, allowing the identification of slight anomalies corresponding to zero-day attacks [147]. The ability of SAEs to learn nonlinear patterns makes them highly suitable for complex intrusion datasets with high feature variability.

Table 4 A comparison of the commonly used datasets for IDS and zero-day threats

Dataset name	Year	Types of attacks (It Contains)	Focus	Features	No. of samples	References
CIRA-CIC-DoHBrw-2020	2022	HTTTPS (Non-DoH and Benign DoH), Malicious DoH	Detect DoH attacks	NA	3,114,791	[137]
CIC-AWS-2018	2018	DoS, Probe, R2L, U2R	Cloud-based attacks	NA	2,111,420	[135]
CICIDS2017	2017	DoS, Probe, R2L, U2R, Reconnaissance	IDS research	80	2,830,743	[128]
UNSW-NB15	2015	DoS, Probe, R2L, U2R, Reconnaissance	IDS for modern networks	49	2,540,044	[134]
NSL-KDD	2009	DoS, Probe, R2L, U2R	Enhanced version of KDD-CUP99 with reduced redundancy	43	125,973	[132, 133]
KDD-CUP99	1999	DoS, Probe, R2L, U2R	Classic IDS benchmark dataset	41	4,898,431	[131, 139]

Table 5 The common attacks in the zero-day attack datasets

Attack	Abbreviation	Description
DoS	Denial-of-Service Attack	A DoS attack is a cyberattack that aims to exhaust a target system's by flooding it with a large volume of traffic from a single source [141]
U2R	User to Root Attack	A User to Root (U2R) attack is when an attacker initially gains access to a system as a regular user and then exploits vulnerabilities to elevate their privileges to the root level. This can be done by leveraging weaknesses in the system's software, configuration, or security practices [142]
U2L	Remote to Local Attack	A Remote-to-Local (R2L) attack is a type of cyberattack that involves exploiting system vulnerabilities to gain unauthorized access to a target system from a remote location without possessing legitimate credentials [142]
Probe	Probing Attack	A probing attack is a cyberattack that involves scanning a network or system to identify vulnerabilities and potential points of entry that can be exploited. This involves sending various requests to the network to gather information about its configuration, software, and services [143]

To enhance feature selection further, hybrid approaches that combine various criteria have gained popularity. A common approach is to combine mutual information and linear correlation analysis to choose informative and non-redundant features [148]. Pareto dominance set approaches to provide a multi-objective method for selecting feature pairs trading off information content and correlation. This ensures that the selected features are helpful for model performance while keeping redundancy minimal [148].

Metaheuristic algorithms have been very promising in efficiently exploring the feature space for optimal subsets. An example of such an algorithm is the Honey Badger Optimization (HBO) algorithm, which mimics the intelligent foraging behavior of honey badgers to choose relevant features while excluding irrelevant or noisy data [149]. Similarly, the opposition-based Northern Goshawk Optimization (NGO) algorithm has also been employed in network IDS to enhance feature relevance and model interpretability. Considering both current and opposite solutions during the search, this method enhances the chances of discovering a globally optimal feature set [146].

Feature selection techniques can also be tailored to specific network security applications. For example, univariate feature selection has been employed to enhance the performance of deep CNNs in botnet attack detection by focusing on the most statistically prominent features [150]. Meanwhile, wrapper-based feature selection approaches are effective in hybrid deep learning models for detecting DDoS attacks in Network Function Virtualization (NFV) environments [151]. These methods evaluate feature subsets using model validation and training, thus ensuring that only the most impactful features are retained for real-time detection applications.

Generalization of IDS is extremely difficult, particularly when dealing with zero-day attacks, which are new and previously unseen [152]. Traditional ML-based IDSs are likely to fail because they have learned from specific sets of data and types of attacks and thus perform poorly when encountering data with different distributions [153]. This is because models cannot recognize network traffic's inherent characteristics correctly, resulting in weak generalization [153]. Different studies work to address these limitations in different ways. For

example, unsupervised learning techniques are explored to enable IDSs to detect anomalies without requiring labeled data [153]. Self-supervised contrastive learning (SSCL) is another method employed to improve the generalization of IDSs through learning basic network traffic features and classification of new, unseen network flows [153]. Meta-learning and FL are also discussed to create powerful and adaptable IDSs to detect known and zero-day attacks in different environments [58].

To solve the problem of generalization, especially for zero-day attacks, hybrid AI-based approaches are proposed [154]. These methods combine ensemble and DL to enhance zero-day exploit detection in IDSs [155]. In addition, anomaly detection methods based on CNNs and regularization are utilized to identify zero-day attacks in IoT systems [156]. Transfer learning techniques are also used to leverage knowledge in other domains to improve the detection of zero-day attacks in specific systems, such as communication-based train control (CBTC) systems [157]. Besides, developing adaptive DL-based IDSs that can learn and adapt to new attacks is significant [152]. These adaptive systems are apt to employ online and active learning methods for the purposes of regularly updating their detection models with new data [158]. Their goal is to create IDSs that excel at detecting known attacks but also possess the capability to generalize to unknown attacks and hence provide robust protection against zero-day exploits [152].

5 Zero-day attack detection based on AI techniques

Integrating AI with cybersecurity is now essential, offering advanced solutions to address escalating cyber threats. AI, powered by ML and DL algorithms, can analyze large-scale datasets to identify emerging threats, enabling the development of proactive security measures. These adaptive systems enhance resilience by continuously learning from emerging risks, significantly advancing cyber defense [159].

The cybersecurity sector is increasingly investigating how AI can be leveraged to detect these hard-to-trace attacks. With its capacity to process extensive datasets, recognize complex patterns, and adjust to emerging threats, AI stands out as a potentially revolutionary tool in combating zero-day attacks [160].

5.1 Related techniques based on ML and DL models' approach

Research on novel attack detection has explored various ML and DL approaches for malware analysis. For example, authors in [161] introduce HERO, a DL approach grounded in the neural algorithmic reasoning (NAR) framework for zero-shot detection of zero-day attacks in unprocessed high-dimensional network traffic. The method involves initially training a neural processor network on a low-dimensional synthetic dataset to simulate anomaly detection, enabling it to learn essential reasoning patterns. Subsequently, encoder and decoder networks are attached to this pre-trained processor and trained on real raw network traffic data, represented as high-dimensional packet byte matrices, to map the raw input into the processor's learned manifold. This separation of feature representation from decision-making allows HERO to effectively process noisy, high-dimensional inputs and generalize to previously unseen attack types without requiring labeled examples.

In another contribution in [162], the authors utilized the extreme learning machine (ELM) as a classifier to improve the performance of IDS in IoT networks. The methodology included data collection, preprocessing, training, and testing phases, with the ELM model implemented

using Python and Google Colab. The dataset was divided for both binary classification (normal vs. attack) and multi-class classification (identifying specific attack types). The approach prioritized handling high-dimensional and unbalanced data to boost detection accuracy and efficiency, and the ELM model's performance was compared with other supervised learning models such as Random Forest (RF), K-Nearest Neighbors (KNN), and Decision Trees (DT).

Further contributions to malware detection include authors in [33], who optimized multiple ML classifiers and utilized feature engineering techniques like PCA and Chi-Square to improve model performance on limited datasets. Another approach [163] involved two DL-based methods for classifying malware families: a pre-trained CNN model and feature extraction via CNN to train traditional classifiers, achieving an accuracy of up to 92% by visualizing malware as images. Such advancements highlight the move beyond signature-based detection. Additionally, [163] presented a hybrid model for intrusion detection that combines autoencoders with RF and XGBoost, enhancing zero-day detection by identifying previously unseen patterns.

Research has also explored various algorithms for detecting anomalies and ransomware, as in [104], where NN and RF were highly effective, though k-means clustering struggled with specificity in limited data diversity. A novel ML model by [164] focused on detecting zero-day ransomware attacks by identifying behavioral indicators without relying on signatures, while [51] introduced a zero-day detection framework that integrates supervised and unsupervised classification techniques, enhancing accuracy by detecting outliers with the K-Means algorithm. These studies, summarized in Table 6, collectively advance zero-day attack detection by incorporating diverse, adaptive methodologies for real-world applicability.

5.2 Related techniques based on the FL model approach:

Recent studies have applied FL to address zero-day cyberattack detection with various innovative frameworks. In [167], a decentralized FL framework called DTFL-Audit is proposed for zero-day attack detection in hospital settings, integrating blockchain to maintain data security and privacy. Transactions within the framework are regulated through smart contracts, and a score-of-acceptance method is introduced to dynamically adjust the accuracy threshold for attack detection. Another study, [168] presents a multi-stage IDS tailored for in-vehicle networks to detect both known and unknown threats. This system employs DL algorithms to capture sequential dependencies in CAN messages, achieving high detection rates while maintaining a lightweight structure suitable for FL deployment. By leveraging FL, the IDS can protect user privacy while utilizing diverse data from various locations and driving environments.

Further addressing zero-day attack detection in decentralized systems, [169] introduces a model that enables clients to customize FL-based IDS models to accommodate diverse data sources, achieving impressive detection accuracy between 95–99%. This approach effectively identifies zero-day attacks by tailoring model performance to specific client requirements. Additionally, an Edge-based FL and blockchain-integrated framework for Intelligent Transportation Systems (ITS) is proposed in [170]. This decentralized system combines edge computing with a reputation mechanism to ensure secure and accurate threat detection, demonstrating high efficacy on the UNSW-NB15 dataset and preserving privacy across distributed ITS environments.

For zero-day detection in medical IoT networks, [58] presents Anoml-Fed-IDS, an IDS specifically for the Internet of Medical Things (IoMT). This system uses biased classifiers

Table 6 The studies based on ML & DL

Reference	Year	Dataset	Methodology	Performance
[161]	2025	BoT-IoT Dataset	HERO, a deep learning framework based on the NAR paradigm for zero-shot detection of zero-day attacks	Accuracy: 0.991 ROC-AUC: 0.990
[162]	2024	NSL-KDD	ELM as a classifier to enhance IDS performance in IoT networks	Accuracy: ELM achieved 99.78% in binary classification ELM achieved 99.90% in multi-class classification
[33]	2024	Custom malware dataset	RF, SVM	Accuracy: 92%, Precision: 90%, Recall: 88%
[163]	2024	Custom malware image dataset	CNN	Accuracy: 94%, False Positive Rate: 5%
[165]	2024	UNSW-NB15	Autoencoders, RF	Detection Rate: 93%, Precision: 89%, Accuracy: 91%
[104]	2024	NSL-KDD, UNSW-NB15	DL (DNN, CNN)	F1-score: 92%, Detection Rate: 90%
[164]	2024	Custom ransomware dataset	Ensemble models (RF + XGBoost)	Precision: 92%, Recall: 90%, F1-score: 91%
[51]	2024	IBM & NSL-KDD	A hybrid approach that integrates both supervised and unsupervised learning techniques Supervised models: DT, RF, KNN and Naive Bayes (NB) Unsupervised model: K-Means clustering for pattern recognition and anomaly detection	Accuracy of 98.4% for the IBM dataset and 96.6% for the NSL-KDD dataset
[166]	2024	Custom malware dataset	One-class classifier	Accuracy: 92%, Precision: 90.38%, Recall: 94% F1-score: 92.16%

Table 7 Studies that are based on FL

Reference	year	Domain	FL approach	ML approach	Performance metrics
[167]	2024	Healthcare 4.0	FL with secure auditing	Deep Reinforcement Learning	Detection Rate: 91%, Accuracy: 92%
[168]	2024	In-Vehicle Networks	Hierarchical Federated Learning (H-FL)	DL, Ensemble Methods	Precision: 93%, Detection Rate: 95%
[169]	2024	Medical IoT	Federated Transfer Learning (FTL)	Transfer Learning, Neural Networks	Accuracy: 92%, Recall: 90%, F1-score: 91%
[170]	2024	Vehicular Edge Computing	FL with Blockchain	DL	Accuracy: 94%, Precision: 92%, Recall: 91%
[58]	2024	IoMT	Meta-Federated Learning	Meta-Learning, FL	Accuracy: 93%, Precision: 91%, Detection Rate: 92%

and agglomerative clustering for local model training, with secure aggregation in FL. A new dataset, M-IoT-Env, was created by merging WUSTL-IIOT-2021 and IoTID20.

to evaluate its performance. The model also addresses class imbalance via SMOTE and employs feature engineering techniques such as RF-RFESCV and PCA, with hyperparameters optimized through HSO and BO-TPE. Designed for real-time use in IoMT, fog, and cloud environments, Anoml-Fed-IDS demonstrates high accuracy and low misclassification rates. These contributions, summarized in Table 7, represent significant advancements in applying FL for robust and privacy-preserving zero-day attack detection.

6 Future directions

The detection of zero-day attacks using ML is rapidly evolving, but several challenges remain, offering promising avenues for future research. As the landscape of cybersecurity threats becomes more complex, the following directions hold significant potential for advancing the detection of zero-day attacks [171].

6.1 FL and privacy-preserving approaches

Centralized ML models can be vulnerable to data breaches and privacy violations, especially when dealing with sensitive data, such as medical records or financial information [172, 173]. FL enables multiple clients to participate in model training by sharing model updates, with a central server rather than raw data [174]. This approach can help detect zero-day attacks in environments where privacy restricts data sharing [167].

FL in cybersecurity research could focus on how FL can be applied across different organizations, each contributing to a global model while keeping their data private. Such models could be trained on attack data from multiple networks without exposing sensitive

information, thereby improving the detection of previously unseen zero-day attacks across distributed systems [175].

Challenges in FL for zero-day detection: One major hurdle is the communication cost and latency in FL. Ensuring efficient and scalable deployment in real-time threat detection scenarios is crucial for future research.

Privacy-preserving ML combining FL with other privacy-preserving techniques like homomorphic encryption or differential privacy could offer additional layers of security [176]. This approach enables security and privacy-preserving ML by training models on encrypted data, improving the detection of zero-day vulnerabilities [173].

6.2 Explainable AI (XAI) for zero-day detection

As ML models become more complex, their decisions often resemble "black-box" outputs, making it difficult for cybersecurity professionals to understand how these models identify zero-day threats [107]. XAI seeks to provide transparency by offering explanations for the predictions made by ML algorithms.

Explainability is crucial in cybersecurity for zero-day attack detection, it is crucial for system administrators to trust and understand the decisions made by AI models. XAI techniques can help in interpreting the reason for a specific activity is flagged as malicious or benign. This increases confidence in automated detection systems, especially when applied to high-stakes scenarios like critical infrastructure protection [177].

Research opportunities in XAI developing methods that provide detailed explanations of decisions made by DL models used in zero-day attack detection are a promising area. Techniques like local interpretable model-agnostic explanations (LIME) or SHapley Additive exPlanations (SHAP) can be further researched to offer more granular insights [178, 179]. Future work should focus on developing XAI techniques that can provide insights into the decision-making process of complex models.

6.3 Adversarial ML and robustness

As ML models become more widely used in detecting zero-day attacks, they become targets of adversarial attacks. Adversaries can craft malicious inputs, known as adversarial examples, to fool the ML models into misclassifying or overlooking attacks. In adversarial attacks in zero-day detection, attackers may create slightly modified versions of malicious traffic that appear benign to an ML model but still exploit vulnerabilities. Research into adversarial learning focuses on making models more robust against such manipulations. One solution is to train models using adversarial examples, a technique known as adversarial training [180].

Robustness and resilience of ML models in future research should explore techniques to make ML models more resilient to adversarial attacks. Methods like input preprocessing, adversarial detection systems, and robust architecture designs (e.g., deep NN combined with anomaly detection) could significantly improve the robustness of zero-day detection systems.

6.4 Real-time detection and scalability

The real-time detection of zero-day attacks remains a significant challenge for current ML-based systems [181]. Zero-day exploits often happen within minutes or seconds, making it imperative for detection systems to operate with minimal latency. To address the scalability

challenges posed by the growth of networks and IoT devices, future research should explore techniques such as distributed and parallel computing, and efficient data processing and feature engineering methods, to optimize the performance of ML-based anomaly detection systems. Edge and cloud computing for real-time detection, leveraging edge computing, can help distribute the computational load by detecting anomalies closer to the source of data generation. Combining edge and cloud-based architecture could offer a scalable solution for real-time detection of zero-day threats, ensuring timely response without overloading central processing units [182].

6.5 Integration of AI with threat intelligence platforms

Integrating AI-powered zero-day attack detection systems with threat intelligence platforms can greatly enhance them. These platforms aggregate data on known vulnerabilities, attack signatures, and potential threats from various sources [121, 170].

AI and threat intelligence synergy by integrating threat intelligence data into ML models, these systems can continuously update and refine their detection capabilities. This integration enables proactive detection, where emerging patterns or behaviors indicative of potential zero-day vulnerabilities, are flagged based on the latest intelligence data [183]. Collaboration between human analysts and AI, combining AI with human expertise, remains a key research area. AI systems can handle vast amounts of data and detect anomalies, while human analysts provide the contextual understanding and decision-making required to assess complex threats. Developing frameworks where AI models assist analysts by prioritizing the most critical alerts and providing insights into potential zero-day vulnerabilities is a future direction worth exploring [184, 185].

6.6 Transfer learning for zero-day attack detection

Transfer learning enables models trained on one task to be adapted to a related task, reducing the required training data. This is especially useful in zero-day attack detection, where labeled data for specific attack types might be scarce or unavailable [87]. Leveraging transfer learning for zero-day detection is a promising research direction [98]. For instance, a model trained on known network intrusions could be adapted to detect novel forms of network-based zero-day exploits [185]. Few-Shot and Zero-Shot Learning: Few-Shot learning, where models learn from minimal data, and Zero-Shot learning, where models generalize to unseen categories, are promising areas for zero-day detection. Research could explore how these techniques can improve the adaptability of ML models to emerging zero-day threats [186].

6.7 FL and poisoning attacks: affect and proposed potential defense strategies

Poisoning attacks in FL represent a significant security concern, especially regarding the integrity and reliability of zero-day attack detection models. FL, designed to enable collaborative model training without direct data sharing, ironically introduces vulnerabilities exploitable by malicious actors [187].

Poisoning attacks in FL involve adversaries injecting malicious data or model updates into the training process to manipulate the global model [188]. Poisoning attacks are two types, which are data poisoning and model poisoning. Data poisoning involves poisoning the training data, and model poisoning involves messing around with the model [189]. The consequences of

zero-day detection models are grim: compromised model accuracy. Poisoning attacks can significantly lower the global model's accuracy, leading to an elevated FNR in detecting new, unseen (zero-day) attacks [190]. This occurs because the model becomes trained on the erroneous patterns from the poisoned data and thus becomes ineffective against novel threats [191]. Backdoor insertion: backdoors can be added by attackers into the model such that it misclassifies specific inputs as normal or attacks based on a pre-specified trigger. It is especially dangerous for zero-day attack detection, as attackers can set up inputs that are capable of evading detection by triggering the backdoor [192]. Model destabilization: byzantine attacks, which form a class of model poisoning attacks, include adversarial clients emitting corrupted model updates, and this can cause the global model to diverge or converge into a sub-optimal model. Destabilizing the model undermines the model's generalization ability as well as its capability to detect zero-day attacks accurately [193, 194]. Evasion of detection: Poisoning attacks can be designed to specifically evade existing detection methods. For instance, a malicious actor can tailor the poisoned data points to look like legitimate data so they would be difficult to detect as attacks using routine anomaly detection methods [195]. Privacy Leakage: Despite FL's attempt to preserve privacy, poisoning attacks can be combined with inference attacks to leak sensitive information. A malicious actor can potentially cause the model to reveal patterns or characteristics of the training data, thereby invading the privacy of the individual clients [196].

Potential defense approaches to mitigate the impact of poisoning attacks against FL-based zero-day detection models include certain defense methods, which are (i) Data Sanitization and preprocessing: implementation of robust data preprocessing techniques can be employed to identify and remove potentially poisoned data samples. These are validation checks, data normalization, and outlier detection to maintain data integrity [197]. (ii) Anomaly detection: Using anomaly detection methods on the model updates submitted by clients can detect corrupted or malicious updates. Techniques such as Isolation Forest and density-based clustering can be utilized to detect anomalies within model weight distributions [196]. (iii) Robust aggregation techniques: Employing robust aggregation techniques can minimize the impact of malicious model updates. Methods like federated median, Krum, and trimmed mean are less susceptible to outliers and can effectively discard poisoned updates [198]. (iv) Adversarial training: Utilizing adversarial training methods during training the global model can make it more resistant to poisoning attacks. This is achieved by generating adversarial examples imitating poisoned data and instructing the model to predict them with high accuracy [199]. (v) Meta-learning-based defenses: Utilizing meta-learning for poisoning attack detection and resistance allows dynamic adjustment of the defense mechanism in sync with the client behavior observed. This can learn new and evolving forms of attacks and apply an enhanced defense [200]. (vi) Autoencoder-Based Latent Space Inspection: approaches like Fed-LSAE utilize autoencoders to scan for latent space representations of model updates and identify anomalies that indicate poisoning attempts. This process can effectively remove the malicious updates using the autoencoder's ability to reconstruct innocuous representations [201]. The Fed-LSAE scheme algorithm can be described by the procedure for PLRs (Penultimate Layer Representations) extraction, computation of Auto-Encoder-based Latent Space Representations, and computation of CKA (Centered Kernel Alignment) Scores for gathering robust global weight. Fed-LSAE mechanism against poisoning attacks. (vii) Formal Logic-Guided FL: Incorporating formal logic into the training process of FL can help preserve the reliability and integrity of the global model. Formal logic-based rules and constraints can be set, and the system can mark and eliminate model updates that fail to follow these rules, thereby applying a strong mechanism against poisoning attacks [202]. (viii) Dynamic client selection: Implementing dynamic client selection techniques can optimize participant

selection and mitigate threats from malicious contributions mitigated. The reliability of the vehicles being evaluated, the technique can improve the accuracy level of the aggregative model and safeguard against potential security attacks [203].

Historical distance detection: avoidance of a FL poisoning attack through historical distance detection [204]. Privacy-Preserving Techniques: Combining defense mechanisms with privacy-preserving techniques such as differential privacy can enable the defense mechanisms to refrain from violating the privacy of the involved clients [205].

6.8 The trade-off between computational cost and detection accuracy

The application of ML to identify zero-day attacks is a critical trade-off between detection precision and computational cost. Zero-day attacks are based on unobserved vulnerabilities, so signature detection is impossible. ML approaches are highly promising, but with the warning that implementation should be careful in being responsive to both computational overhead and performance needs [206].

The inherent challenge of applying ML to zero-day attack detection is designing models that generalize well across unknown attack patterns [207]. There are various approaches with varying computational requirements and detection capability: supervised learning: supervised approaches require labeled training sets of good and bad traffic to train a model [208]. While they can achieve high rates of detection, the need for large quantities of labeled data, especially for new and novel forms of attack, turns out to be computationally expensive and time-intensive [209]. Unsupervised learning: These approaches attempt to identify outliers in network traffic without prior knowledge of the attack signatures [210]. Methods like clustering and autoencoders can feel anomalies from the norm and potentially identify zero-day attacks. Unsupervised approaches are computationally less costly than supervised learning but can also have poorer FPR [211]. Semi-supervised learning: These techniques combine labeled and unlabeled data to increase detection accuracy [211]. They can leverage a limited amount of labeled data to govern the learning process on a large set of unlabeled data to offer a balance between detection performance and computational cost. FL enables simultaneous model training on multiple devices or organizations without sharing sensitive information [212]. Detection accuracy can be improved with the help of FL by leveraging heterogeneous datasets with privacy preservation. Overhead of communication along with the potential of biased data in FL systems adds to computational complexity.

Big datasets, particularly volumetric DDoS attack-related ones, present severe challenges to model efficiency, scalability, and real-time detection [213]. The sizes of these datasets necessitate more computationally expensive algorithms, which influence model training and inference time [214]. In the context of real-time detection, this translates into demanding remarkably optimized models that can process gargantuan data streams with minimal latency [215]. Scalability is required when there is an increase in data, and there must be a distribution of the computation load through mechanisms like FL to preserve detection accuracy [216]. Further, the difficulties that are created by large-volume data can exacerbate issues like class imbalance, wherein attack data are far less in quantity than normal traffic, leading to biased models and poor detection [217]. Mitigation these issues necessitates innovative approaches to model development, feature detection, and distributed training paradigms to offer timely and efficient defense against volumetric DDoS attacks [218].

6.9 Zero trust architecture framework

Zero Trust Architecture (ZTA) is an approach to security founded on the "never trust, always verify" premise and follows through on it by removing implicit trust and reliance on identity verification, enhancing cybersecurity [219]. In contrast to the conventional perimeter-centric security models of trust within the network, ZTA demands ongoing authentication and approval of all its end users and devices, irrespective of where they are (on-premises or elsewhere) [220]. This model is vital in today's environments where the network perimeter is becoming more obscure due to cloud computing, mobility, and remote connectivity [221]. Strong authentication, least privilege access control, micro-segmentation, and ongoing monitoring are some essential elements of ZTA [222]. By applying ZTA, organizations hope to reduce the effect of breaches through limited exposure of the attack surface and halting lateral movement in the entire network [223].

7 Conclusion

The integration of ML into cybersecurity is revolutionizing threat detection by enabling proactive, intelligent, and adaptive responses to zero-day attacks. Key areas such as FL, XAI, and adversarial robustness are essential for improving the detection systems' accuracy, scalability, and security. These approaches allow real-time processing and learning from evolving threats, helping organizations stay ahead in an ever-changing cyber landscape. While the potential is vast, addressing the current limitations of ML and strengthening its resilience against adversarial attacks remains critical for long-term effectiveness. The review also emphasizes the unique potential of FL in cybersecurity. FL, with its decentralized data learning approach, supports privacy-preserving and secure on-device AI, making it a suitable solution for sensitive data environments. By analyzing its application in various fields, such as finance and computer vision, the paper highlights the potential of FL and underscores the challenges that need to be overcome to realize its full potential in real-time cybersecurity scenarios. This review offers future directions to enhance FL's role in threat detection, paving the way for more resilient defenses against emerging cyber threats.

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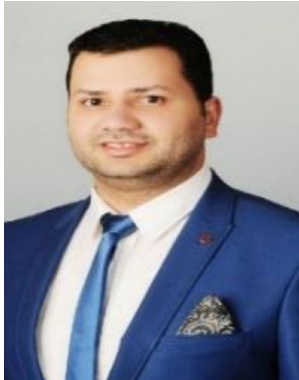
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